

GREGORY GOLONKA

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Education

University of Notre Dame — B.S. Aerospace Engineering, May 2025

- GPA: 3.53
 - Relevant Coursework: Thermodynamics, Heat Transfer, Fluid Mechanics, Compressible Aerodynamics, Gas Turbines & Propulsion
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Technical Skills

- Programming & Simulation: Python, MATLAB, Fortran, C++, HTML, Next.js
 - CAD & Design: SolidWorks, Autodesk Fusion 360, XFLR5, XFoil
 - Tools: CHEMKIN-II, LSODE, ONELAB/Gmsh, Wind Tunnel Instrumentation
 - Engineering: Propulsion Systems, Flight Dynamics, Experimental Aerodynamics
 - Soft Skills: Technical documentation, leadership, project management
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Projects & Research

Senior Design — Radio-Controlled Aircraft

- Designed a radio-controlled airplane to complete a specific mission, focusing on aerodynamic optimization and structural integrity.
- Utilized XFLR5 and SOLIDWORKS CFD to analyze and refine the base model through iterative simulations.
- Led budgeting, logistics, and construction; managed CAD-guided assembly and sourced materials balancing cost, weight, and durability.

Undergraduate Research — Shock Wave Structure Analysis

- Conducted independent study into viscous shock structures under Dr. Joseph M. Powers.
- Investigated dissociation of diatomic molecules under extreme shock conditions using 1D Navier-Stokes equations.
- Advanced understanding of propulsion and detonation modeling by linking molecular dynamics to continuum-scale physics.

Computing Project — Heat Transfer in Food Systems

- Developed a numerical model to simulate transient heat conduction in a steak during cooking.
- Accounted for variable oven temperatures to optimize char formation while maintaining sub-medium-rare interior.
- Created predictive model for cook-time estimation under uncertain parameters.

Personal Project — Portfolio Website -- <https://gtgonka.com/>

- Built first personal website using JavaScript, HTML, Next.js, and CSS.
- Collaborated with brother (software engineer) to refine design and functionality.
- Strengthened skills in frontend development and project self-learning.

Experimental Aerodynamics — Wind Tunnel Studies

- Calibrated pressure transducers and collected digital data from analog signals.
- Analyzed airfoil circulation, pressure distribution, and lift/drag characteristics of finite wings.
- Developed strong foundation in experimental methods and aerodynamic force analysis.

Gas Turbines & Propulsion — Williams F-107 Engine Redesign

- Performed thermodynamic analysis of Williams F-107 turbofan for improved efficiency.
 - Redesigned engine evaluated with mission simulation code covering ascent, cruise, and maneuvering phases.
 - Quantified fuel consumption and proposed redesign improving mission-specific performance.
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Work Experience

Movie Theater Projectionist — Capitol Theater (2023–Present)

- Troubleshoot and maintain digital projection systems to ensure uptime of events.

Dorm Supervisor — Dunne Hall, University of Notre Dame (2022–2023)

- Managed front desk operations and supported in-dorm services, ensuring resident needs and safety were met.

Landscape Technician — C. Michaud Landscaping (2021–2024)

- Coordinated team operations across 150+ weekly properties, maintaining high service quality.